

Constraint Closure Coherence Time Theorem (CCT): A Unified Dynamical Framework for Lifetime in Living and Non-Living Systems

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Abstract

The observation that lifetime correlates positively with structural complexity spans the entire empirical spectrum — from molecular complexes to multicellular organisms to stars and galaxies. Yet existing theories explain this pattern only within their respective domains: Kleiber's law and evolutionary senescence theory are confined to biology; fracture mechanics addresses only materials; the free energy principle does not provide a temporal dynamics of persistence. Here I propose the Constraint Closure Coherence Time Theorem (CCT), derived from the generative dynamics framework of the Holographic Dual-Evolution Theory (HDET). The theorem states that for any closed constraint closure — a self-sustaining system in which higher-level inertial structures constrain lower-level observer activities while lower-level activities maintain higher-level structures — lifetime T_{life} is strictly determined by the decoherence rate Γ of its individual Θ -potential:

$$T_{\text{life}} = \Theta_0 / \Gamma(N)$$

where N is the nesting depth of the constraint closure and Γ is a monotonically decreasing function of N . The theorem yields two empirically testable branches: in living systems, $\Gamma_{\text{bio}}(N) = \Gamma_0 e^{(-\alpha N)} + \gamma_0 / N$ (frequency reduction for life extension); in non-living systems, $\Gamma_{\text{phys}}(N) = \Gamma_0 e^{(-\lambda N)}$ (barrier elevation for lifespan preservation). Cross-scale validation — from van der Waals complexes (10^{-12} s) to galaxies (10^{18} s) — shows consistent quantitative agreement across seven orders of magnitude. Three experimentally testable predictions are proposed, including convergent total heartbeat counts across mammals ($\sim 10^9$), ultra-long room-temperature lifetime for $N \geq 6$ covalent structures ($> 10^9$ years), and a knowledge entropy criterion for civilizational persistence. CCT unifies lifespan across the organic–inorganic divide under a single conservation principle: the total

transformation of individual Θ -potential is conserved; nesting depth determines its dissipation rate.

Keywords: constraint closure; decoherence rate; nesting depth; lifespan; living-nonliving unification; generative dynamics; Holographic Dual-Evolution Theory

1. Introduction: The Cross-Scale Puzzle of Lifetime

1.1 The Ubiquitous Pattern

From molecular complexes to galaxies, a striking pattern recurs across all scales of organized matter: systems with greater structural complexity tend to persist longer. Diamonds (covalent crystals) outlast van der Waals complexes; multicellular organisms outlive unicellular ones; galaxies persist longer than their constituent stars (see Table 1 for representative values). This positive correlation between what may be called "structural depth" and "temporal endurance" is so widely observed that it risks being taken for granted.

Yet its ubiquity masks a profound theoretical lacuna: no existing theory explains this correlation across the living–nonliving divide.

1.2 The Fragmentation of Existing Explanations

Current theories that address lifespan or persistence are confined to their respective domains:

Kleiber's law [1] and its extensions (metabolic rate $\propto M^{3/4}$, lifespan $\propto M^{1/4}$) explain why larger mammals tend to live longer — but have nothing to say about why diamonds outlast graphite, or why galaxies persist for billions of years. The law is explicitly biological, its metabolic basis inapplicable to inert matter.

Evolutionary senescence theory [2, 3] explains aging through declining selection pressure with age — but only for organisms under natural selection. It cannot explain why identical chemical bonds persist longer in deeper-embedded configurations, or why a star's lifetime depends on its mass stratification rather than any selection process.

Fracture mechanics and materials science [4] explain creep lifetime through activation energy Q via $t_f \propto e^{(Q/RT)}$ — but these are phenomenological descriptions of specific mechanisms (dislocation climb, diffusion) rather than a unified principle applicable to biological metabolism.

The Free Energy Principle (FEP) [5] provides a unified account of self-organization across scales but focuses on "belief updating" and variational free energy minimization — it does not specify how long a system can persist given its structural configuration.

In short: lifespan is treated as a biological variable, while persistence is treated as a physical variable — and the two are never brought under a single conceptual roof.

1.3 The Need for a Unified Framework

A unified framework for lifetime would need to satisfy three conditions:

1. **Scale-transcendence:** It must apply equally to molecules, organisms, stars, and galaxies.
2. **Mechanism-independence:** It must be expressible in terms of structural rather than material-specific parameters.
3. **Testability:** It must yield empirically verifiable predictions at multiple scales.

The Constraint Closure Coherence Time Theorem (CCT), derived from the generative dynamics of the Holographic Dual-Evolution Theory (HDET) [6], is proposed to meet these conditions. The theorem's central claim is that lifetime is not a biological property but a universal temporal attribute of constraint closures — any self-sustaining system in which higher-level structures constrain lower-level activities while lower-level activities in turn maintain the higher-level structures.

The CCT states that for any such constraint closure:

$$T_{\text{life}} = \Theta_0 / \Gamma(N)$$

where Θ_0 is the system's initial individual Θ -potential (the "grammatical capital" allocated at its individuation), and $\Gamma(N)$ is the decoherence rate — the instantaneous rate at which this Θ -potential dissipates into the environment — which is a monotonically decreasing function of the nesting depth N .

The theorem immediately yields two distinct branches:

- **Living systems** (active maintenance, $\Gamma_{\text{bio}} \propto 1/N$ in the deep nesting limit): "frequency reduction for life extension"

- **Non-living systems** (passive resistance, $\Gamma_{\text{phys}} \propto e^{(-\lambda N)}$): "barrier elevation for lifespan preservation"

Both branches share the same underlying ontological constraint: conservation of individual Θ -potential transformation.

1.4 A Critical Clarification: Heart Rate as a Reduced Proxy

A common misinterpretation of the living branch is that "slower heart rate guarantees longer life." This is false. Heart rate is only a proxy for decoherence rate under specific conditions — specifically, when emotional and metabolic energy is directed outward as physical work (benign frequency reduction). When emotional energy undergoes reverse transformation — being suppressed, internalized, or converted into chronic neuroendocrine stress without external physical expression — heart rate may remain low or even decrease, while the effective decoherence rate Γ_{eff} increases sharply due to internal constraint-closure wear [6, 7].

Thus the CCT's living branch distinguishes:

- **Benign deceleration:** Γ_{eff} decreases with N because all levels of O -activity (including neuroendocrine) are genuinely reduced.
- **Malignant deceleration:** heart rate decreases but Γ_{eff} increases due to internal vibrational stress — the constraint closure's internal O/I coupling is being eroded from within.

This is why the CCT theorem's primary criterion is $\Gamma(N)$, not heart rate. Total heartbeat count convergence is a B-level empirical regularity, not the theorem's foundational claim. The theorem's A-level claim is that $\Gamma(N)$ — the generalized decoherence rate — determines lifetime.

1.5 Structure of the Paper

Section 2 introduces the necessary theoretical framework: constraint closures, nesting depth, individual Θ -potential, and decoherence rate. Section 3 presents the CCT theorem in full mathematical form. Sections 4 and 5 derive the two branches for living and non-living systems respectively, with empirical validation. Section 6 provides cross-scale unification from molecules to galaxies, with explicit methodological caveats. Section 7 compares CCT with existing theories. Section 8 proposes three experimentally testable predictions. Section 9 concludes with implications and limitations.

2. Theoretical Framework: Constraint Closures and Nesting Depth

2.1 The Generative Dynamics Framework

The CCT theorem is derived from the HDET generative dynamics framework [6], whose core axioms are briefly summarized here for accessibility.

Axiom I (Θ -conservation): There exists an ultimate, undifferentiated, constant-total potential substrate — the cosmic potential Θ . It is the source and destination of all realization events. $\Theta_{\text{total}} = \text{constant}$.

Axiom II (Dialectical symmetry breaking): Within Θ 's ground state, intrinsic dialectical tension drives spontaneous symmetry breaking, producing the fundamental dynamic gradient $A \propto |O - I|$ which drives all evolutionary processes.

Axiom III (O/I dialectical realization): In any realization event, localized Θ differentiates into two dialectically complementary modalities — Observer Energy (O, representing exploration, expression, outward-directed activity) and Inherent Energy (I, representing stability, structure, memory, inward-directed preservation). In any closed system, $O + I = \Theta_{\text{local}}$.

Axiom IV (Dynamical Scale Effect, DSE): All observable phenomena and attributes are co-generated, relational manifestations when an observer system and an observed system enter symmetrical coupling based on their respective O/I structures.

The key implication of Axiom III is that every localized realization event has a fixed "budget" of Θ -potential — and this budget is what any system must spend to persist.

2.2 Constraint Closures

A constraint closure is a system with a closed causal loop in which:

- Higher-level I-structures (inertial structures) constrain the possible space of lower-level O-activities (observer activities)
- Lower-level O-activities maintain the higher-level I-structures through their collective operations

If either direction fails — if higher-level constraints lose their grip, or if lower-level activities no longer sustain the higher-level structures — the constraint closure dissolves.

Examples across scales:

Scale	System	Higher-level I-structure	Lower-level O-activity
Molecular	Covalent bond	Shared electron cloud	Nuclear vibrations
Biological	Organism	Organ/system architecture	Cellular metabolism
Ecological	Ecosystem	Trophic network structure	Species interactions
Astrophysical	Star	Gravitational stratification	Nuclear fusion
Astrophysical	Galaxy	Dark matter halo potential	Stellar orbital motions

A crucial semantic clarification [6]: The term "closed" in "closed constraint closure" refers exclusively to the closure of the causal loop — the O/I cycle maintains itself without external instructions — and not to thermodynamic isolation. Living systems import free energy and export entropy; the import pathways and transformation rules (I-structures) themselves are not directly altered by external information. The constraint closure is closed in grammar, not in energy.

2.3 Nesting Depth N

A constraint closure may contain sub-closures recursively embedded within it. The nesting depth N is the number of such embedded hierarchical levels.

Operational definition: For any constraint closure, identify all distinguishable structural levels at which higher-level constraints operate on lower-level activities. N is the count of such levels.

Examples:

- A van der Waals complex: $N \approx 1$ (single intermolecular interaction)
- A hydrogen-bonded network: $N \approx 2$ (donor–acceptor pairing in network)
- A covalent crystal (diamond): $N \approx 5\text{--}6$ (C–C bonds, tetrahedral coordination, crystal lattice, grain structure, macroscopic domain)
- A mammal: $N \approx 4\text{--}6$ (cell → tissue → organ → system → organism, with endocrine/neural regulatory layers)
- A Sun-like star: $N \approx 4\text{--}5$ (core → radiative zone → convective zone → photosphere → corona)
- A galaxy: $N \approx 6\text{--}7$ (core → bulge → disk → spiral arms → dark halo → satellite system)

Important limitation: The above comparisons across different ontological levels (chemistry, biology, astrophysics) are presented as statistical equivalence rather than dynamical isomorphism. The mechanisms differ fundamentally. The empirical observation is that systems with similar N values fall into similar lifetime ranges

regardless of mechanism — which is itself informative, but not a claim that the mechanisms are the same.

2.4 Individual Θ -Potential Θ_0

At the moment of a constraint closure's individuation (birth of an organism, formation of a molecule, gravitational collapse of a stellar core), it receives a finite allocation of the cosmic potential Θ . This allocation is the individual Θ -potential Θ_0 .

Θ_0 is neither energy nor information in the physical sense. It is the "grammatical capital" that enables the system to organize and transform energy and information according to its internal generative grammar. According to the LEEC (Life Energy Equivalence Conservation) theorem [6], all living systems receive equal Θ_0 at their individuation — though this initial symmetry is subsequently broken by differential nesting depth.

2.5 Decoherence Rate Γ

The decoherence rate Γ is the instantaneous rate at which a system's Θ -potential dissipates irreversibly into the environment.

Its physical counterparts:

- In living systems: metabolic rate (the rate at which free energy is consumed and dissipated)
- In molecular systems: vibrational relaxation rate or tunneling rate
- In astrophysical systems: radiative energy loss rate or gravitational wave damping rate

The decoherence rate is not an intrinsic constant but a function of the system's nesting depth and internal O/I configuration:

$$\Gamma = \Gamma_0 \cdot e^{(-\lambda N)} \cdot e^{(E_a / (k_B \cdot T_{env}))}$$

where Γ_0 is the base decoherence rate for a system with minimal nesting, $\lambda > 0$ is the barrier-elevation coefficient per added nesting level, E_a is the activation energy for escape from the constraint, k_B is Boltzmann's constant, and T_{env} is the environmental temperature.

Critical extension: The effective decoherence rate Γ_{eff} in living systems is not reducible to the physical metabolic rate alone. When emotional energy undergoes reverse transformation — suppression, internalization, or chronic stress without

external physical expression — the internal O/I coupling undergoes accelerated decoherence. In such cases:

$$\Gamma_{\text{eff}} = \Gamma_{\text{physical}} + \Gamma_{\text{internal_stress}}$$

where $\Gamma_{\text{internal_stress}} = \Gamma_0 \cdot e^{(\beta \cdot |\Lambda_{\text{suppression}}|)}$. When $\Lambda \ll 0$ (I-stiffness, emotional suppression), internal dissipation increases exponentially even as physical heart rate decreases [7]. This explains the clinically observed phenomenon of "calm but accelerated aging" — the constraint closure's internal coherence is being eroded from within while the physical pump appears to run slowly.

The derivation of the base form proceeds in three steps [6]:

1. From Axioms II and III, the O/I symmetry-breaking rate is modulated by I-structural complexity, of which N is a direct measure.
2. From the Recursive Nesting Principle (RNP), each additional nesting level introduces an impedance barrier between the system's phase space and environmental perturbations, reducing tunneling probability by $e^{(-\lambda)}$ per level.
3. From Arrhenius kinetics, thermal activation contributes the factor $e^{(-E_a / (k_B \cdot T))}$.

3. CCT Theorem: Mathematical Formulation

3.1 Theorem Statement

Theorem (Constraint Closure Coherence Time Theorem, CCT):

For any closed constraint closure, the coherence time during which it maintains its O/I symmetric coupling — that is, its lifetime T_{life} — is strictly inversely proportional to the instantaneous decoherence rate Γ of its individual Θ -potential:

$$T_{\text{life}} = \Theta_0 / \Gamma(N) \cdots (1)$$

where Θ_0 is the individual Θ -potential allocated at the system's individuation, and $\Gamma(N)$ is the decoherence rate as a function of nesting depth N .

Corollary: Increasing nesting depth N is equivalent to adding impedance along the Θ -dissipation path. Greater impedance means less Θ escapes per unit time, hence longer coherence time.

3.2 Θ -Potential Dissipation Dynamics

The time evolution of the system's individual Θ -potential follows a first-order linear dissipation equation [6]:

$$d\Theta/dt = -\Gamma \cdot \Theta \cdots (2)$$

Integrating:

$$\Theta(t) = \Theta_0 \cdot e^{(-\Gamma t)} \cdots (3)$$

When $\Theta(t)$ falls below a critical threshold Θ_{crit} , the system can no longer maintain O/I coupling coherence — the constraint closure decoheres (dies, dissociates, collapses). Lifetime T_{life} is defined as the time at which $\Theta(T_{\text{life}}) = \Theta_{\text{crit}}$:

$$T_{\text{life}} = (1/\Gamma) \cdot \ln(\Theta_0 / \Theta_{\text{crit}}) \cdots (4)$$

For order-of-magnitude estimation, if $\ln(\Theta_0 / \Theta_{\text{crit}})$ is approximately constant across scales (range $10^0 - 10^2$, with an estimated error of ± 2 log units), then:

$$T_{\text{life}} \propto 1/\Gamma \propto (\Theta_0 / \Gamma) \cdot \text{const} \cdots (5)$$

which yields the simplified form in equation (1).

3.3 The Status of Θ_{crit}

A crucial clarification: Θ_{crit} is not an energy threshold but a topological criterion — it is the "minimum computational redundancy required for closed-loop feedback." It is the minimal "grammatical integrity" needed to maintain O/I coupling. This is scale-invariant in principle because all constraint closures share the same generative grammar (Axiom I). However, the cross-scale constant assumption for $\ln(\Theta_0 / \Theta_{\text{crit}})$ is currently at B-level precision (directional criterion, error about ± 2 log units). A-level quantification requires independent empirical calibration across scales [6].

3.4 Projection onto $\Lambda - C_{\Omega}$ Phase Space

In the HDET diagnostic framework, the $\Lambda - C_{\Omega}$ phase diagram contains families of isochronic curves (constant-lifetime contours). Systems on the same isochronic curve can achieve identical lifetime through radically different strategies:

- **High-frequency, low-barrier strategy** (high Λ , low C_Ω): Rapid metabolism, high exploration, short cycle time — lifetime limited by rapid Θ_0 dissipation.
- **Low-frequency, high-barrier strategy** (low Λ , high C_Ω): Slow rhythm, high I-structural redundancy — lifetime extended by reduced decoherence.

These two strategies are exactly the phase-space expressions of the two CCT branches developed below.

4. Branch I: Living Systems — Frequency Reduction for Life Extension

4.1 The Distinctive Characteristic of Living Systems

Living systems differ from non-living systems in possessing autonomous O-activity — the capacity to actively import free energy, maintain internal low entropy, and repair I-structural damage. This capacity, however, comes at a cost: the I-structure must continuously sustain wear from O-activity. Living systems therefore face a lifetime constraint not of "passive resistance to environmental erosion" but of "active management of self-inflicted wear."

4.2 Derivation of the Frequency Reduction Branch

In living systems, the activation energy E_a in equation (2) manifests as the metabolic threshold required to maintain I-structural integrity. Each additional nesting level (e.g., tissue → organ → system) requires additional maintenance energy. However, because living I-structures have active repair properties — the repair mechanism creates "return flow" in the dissipation path — the decoherence rate is not a simple exponential function of N but has two terms [6]:

$$\Gamma_{\text{bio}}(N) = \Gamma_0 \cdot e^{(-\alpha N)} + \gamma_0 / N \cdots (6)$$

where:

- $\Gamma_0 \cdot e^{(-\alpha N)}$: passive barrier term (exponential attenuation, dominant at small N)
- γ_0 / N : active frequency reduction term (inverse attenuation, dominant at large N)
- $\alpha > 0$: passive barrier coefficient
- γ_0 : base dissipation rate for a single-level constraint closure

Boundary validation: As $N \rightarrow 1$, equation (6) naturally reduces to a single-cell metabolic-rate–lifespan inverse relationship. For yeast or *E. coli*, post-nutrient-depletion survival is determined by residual metabolic rate (γ_0), not by structural nesting. At this boundary, $\Gamma_0 \cdot e^{(-\alpha)}$ and γ_0 compete at similar magnitude — exactly the microbial physiology observation.

Substituting equation (6) into (1):

$$T_{\text{life}} \propto \Theta_0 / (\Gamma_0 \cdot e^{(-\alpha N)} + \gamma_0 / N) \dots (7)$$

In the large-N limit (mammalian deep nesting), $T_{\text{life}} \propto N \cdot (\Theta_0 / \gamma_0)$.

Interpretation: Each additional nesting level forces a proportional reduction in the macroscopic master rhythm (heart rate, respiratory rate), spreading Θ_0 release across longer time scales.

4.3 Distinguishing Benign vs. Malignant Deceleration

A critical clarification is required here. The "frequency reduction" in the large-N limit refers to a genuine reduction in overall O-activity amplitude across all levels, including neuroendocrine and emotional dimensions. However, clinical and physiological observations show that heart rate can decrease while effective aging accelerates — a phenomenon observed in chronic depression, emotional suppression, and "frozen" stress responses [7].

In HDET terms:

- **Benign deceleration:** The entire O/I system reduces its activity amplitude coherently. Heart rate decreases, but so do neuroendocrine stress markers, inflammation, and internal vibrational noise. The system enters a "deep rest" state ($\Lambda \approx 0$, low Γ_{eff}). This is the tortoise/elegant strategy.
- **Malignant deceleration:** Physical heart rate decreases (via vagal tone or pathological bradycardia) but emotional and cognitive O-activity is internally folded — suppressed, not expressed. This suppression converts O-energy into high-frequency internal vibrations: cortisol elevation, sympathetic nervous system dysregulation, inflammatory cytokine release. The physical pump runs slowly, but the internal constraint closure is being eroded by internal entropy generation. Γ_{eff} increases even as heart rate decreases.

This distinction is formalized as:

$$\Gamma_{\text{eff}} = \Gamma_{\text{pump}} + \Gamma_{\text{internal}}$$

where Γ_{pump} scales with heart rate (physical wear) and Γ_{internal} scales with $\exp(\beta \cdot |\Lambda_{\text{suppression}}|)$. When emotional energy is suppressed, $\Lambda \ll 0$ (I-stiffness), and Γ_{internal} can dominate even when Γ_{pump} is low.

This is why the CCT's primary criterion is $\Gamma(N)$ — the generalized decoherence rate — not heart rate. Heart rate is only a B-level empirical proxy, valid only under the condition that O-activity is directed outward (benign frequency reduction). When O-activity is internally folded, the proxy fails.

4.4 Empirical Validation: Cross-Species Heartbeat – Lifespan Data

The living branch predicts that total heartbeat count — heart rate multiplied by lifespan — should converge to a constant across mammals (except for human medical intervention), provided the system is in benign deceleration:

Species	Mass (kg)	Heart Rate (bpm)	Max Lifespan (yr)	Total Heartbeats ($\times 10^9$)
Mouse	0.03	600	3	0.95
Rat	0.3	350	5	0.92
Rabbit	2	200	8	0.84
Dog	15	90	20	0.95
Human	70	72	85	3.2
Elephant	4000	30	70	1.1

Data sources: [1][8][9]

All non-human mammals converge to $\sim 1 \times 10^9$ total heartbeats. This is precisely the large-N limiting expression of equation (7) — $T_{\text{life}} \times \text{heart rate} = \Theta_0 / \gamma_0$ — with the human outlier (due to modern medicine) demonstrating the conditional nature of the predicted constant.

The Blue Whale Case: The blue whale's total heartbeat count ($\sim 8.4 \times 10^7$) appears to deviate below the 10^9 benchmark. This is not a challenge to the theorem but a verification of a sub-effect [10]: deep-sea low temperature reduces the base decoherence rate Γ_0 via the Arrhenius factor $e^{-(E_a/(k_B \cdot T_{\text{env}}))}$ in equation (2), allowing comparable lifespan at lower total heartbeat counts. The blue whale data empirically demonstrates environmental temperature modulation of Γ_0 .

The Depression/Suppression Case: Chronic emotional suppression or depression may present with normal or low heart rate but accelerated biological aging. This is not a challenge to the CCT either — it is a verification of the Γ_{internal} term. When $\Lambda_{\text{suppression}}$ is large, $\Gamma_{\text{eff}} = \Gamma_{\text{pump}} + \Gamma_{\text{internal}}$ increases even while Γ_{pump} is

low, shortening lifetime despite the slow heart rate. This is the formal explanation for why "he who dies of a broken heart may have a slow pulse."

5. Branch II: Non-Living Systems — Barrier Elevation for Lifespan Preservation

5.1 The Distinctive Characteristic of Non-Living Systems

Non-living systems lack autonomous O-activity. They do not actively repair I-structural damage. Their only mechanism for maintaining coherence is passive resistance to environmental erosion — raising barriers to reduce the probability of decoherence-inducing perturbations.

5.2 Derivation of the Barrier Elevation Branch

In non-living systems, E_a manifests as bond energy (molecular scale) or gravitational potential depth (astrophysical scale). Each additional nesting level (single → double → triple covalent bond; van der Waals → hydrogen bond → covalent network) increases barrier height.

Setting $E_a = \lambda N \cdot k_B \cdot T_0$ (where T_0 is reference temperature, λ a dimensionless coefficient) into equation (2):

$$\Gamma_{\text{phys}}(N) = \Gamma_0 \cdot e^{(-\lambda N)} \cdots (8)$$

Substituting into equation (1):

$$T_{\text{life}} \propto (\Theta_0 / \Gamma_0) \cdot e^{(\lambda N)} \cdots (9)$$

Interpretation: Each additional nesting level produces exponential decoherence reduction and exponential lifetime increase. This is why diamond (deep covalent nesting) persists for billions of years at room temperature while van der Waals complexes dissociate in picoseconds.

Physical mechanism: The exponential dependence arises from quantum tunneling's exponential sensitivity to barrier width. Each added nesting level increases the barrier width L (or height), giving tunneling probability $\propto e^{(-2\kappa L)}$, hence exponential lifetime scaling.

5.3 Empirical Validation: Bond Type and Molecular Lifetime

Molecular System	N (Representative)	Bond Energy (kJ/mol)	Room-Temp Lifetime	log ₁₀ Lifetime (s)
van der Waals complex	1	~0.1 – 1	10 ⁻¹² s	-12
Hydrogen bond network	2	~5 – 30	10 ⁻⁹ – 10 ⁻⁶ s	-9 to -6
Single covalent (C – C)	3	~350	years – centuries	7 – 9
Double covalent (C=C)	4	~610	centuries – millennia	9 – 11
Triple covalent (C≡C)	5	~840	millennia – Myr	11 – 14
Covalent crystal (diamond)	6	~348 (per bond)	(per > 10 ⁹ yr	15

Thermal dissociation kinetics data synthesized from [4][11]

log₁₀ (lifetime) increases linearly with N — this is the direct empirical expression of ln T_{life} ∝ N from equation (9).

6. Cross-Scale Unification and the Methodological Boundary of N Comparison

6.1 Unified Scale Table

The CCT theorem can be evaluated across thirty orders of magnitude [6]:

System Type	Nesting Depth N	Γ (s ⁻¹)	Magnitude T _{life} (s)	Magnitude log ₁₀ T _{life}
van der Waals complex	1	10 ¹²	10 ⁻¹²	-12
Covalent molecule	2	10 ⁸	10 ⁻⁸	-8

System Type	Nesting Depth N	Γ (s ⁻¹)	Magnitude T_life (s)	Magnitude log ₁₀ T_life
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(H₂)

Diamond	5	10 ⁻¹⁵	10 ¹⁵	15
Mouse (benign)	4	10 ⁻²	10 ⁷	7
Elephant (benign)	6	10 ⁻⁴	10 ⁸	8
Sun-like star	5	10 ⁻¹⁵	10 ¹⁵	15
Galaxy	7	10 ⁻¹⁸	10 ¹⁸	18

All values are B-level order-of-magnitude estimates (error ±1 log unit) [6].

6.2 Critical Interpretation: Statistical Equivalence, Not Dynamical Isomorphism

A Sun-like star and diamond share $N \approx 5$ and $\log_{10}$ lifetime ≈ 15 . This does not mean their decoherence mechanisms are identical. The star's Γ is determined by nuclear reaction cross-section (dynamic balance: gravity inward, radiation outward). Diamond's Γ is determined by bond energy (static balance: electromagnetic forces maintaining lattice).

However, in phase space, their effective potential well depths happen to fall in the same magnitude range. The CCT theorem reveals: regardless of mechanism, nesting depth N is the dominant determinant of lifetime; specific physical processes are compressed into the coefficient Γ_0 .

This correspondence is characterized as "statistical equivalence" rather than "dynamical isomorphism."

6.3 The Scaling-Law Limitation on Cross-Scale N Comparison

A crucial methodological boundary must be explicitly stated: direct numerical comparison of N values across fundamentally different ontological levels (quantum, chemical, biological, social) currently lacks a unified scaling-law foundation.

One nesting level in an atomic shell, one nesting level in biological tissue, and one nesting level in gravitational stratification are expressions of the same structural grammar under different boundary conditions — but they differ by orders of magnitude in physical energy scale, spatial dimension, and information throughput. Comparing them directly is analogous to comparing "the number of movements in a symphony" with "the number of floors in a building" — formal isomorphism obtains, but dimensional normalization is absent.

Therefore, the cross-scale N comparisons in this paper follow these constraints:

1. **Within-level comparison (valid):** Within a single ontological level (e.g., between mammals, between covalent molecules), differences in N have clear dynamical meaning — deeper nesting implies slower dissipation, longer lifetime.
2. **Cross-level comparison (heuristic):** Similar N values across levels indicate similar effective potential well depth magnitudes — but do not imply mechanism or energy-scale equivalence.
3. **Normalization (future direction):** Strict cross-level N comparability would require a normalized nesting index $N_{\text{norm}} = N / N_{\text{max}}(\text{scale type})$, where N_{max} is the maximum possible nesting depth for that scale type under current physical limits. This would make N_{norm} a unitless number between 0 and 1. At present, N_{norm} has not been empirically calibrated.

This boundary condition does not weaken the CCT theorem — within any given scale type, N remains a strictly monotonic determinant of lifetime. The cross-scale table's value lies in demonstrating "the same grammar projected onto different substrates," not in claiming "nesting layers at one level can be directly added to or compared with those at another level."

7. Comparison with Existing Theories

7.1 Kleiber's Law and CCT

Kleiber's law $B \propto M^{(3/4)}$ (metabolic rate to body weight) and the lifespan–weight relationship $T \propto M^{(1/4)}$ implicitly state $B \times T \propto M$ — total metabolic consumption scales with mass [1].

The CCT living branch predicts $T_{\text{life}} \propto N \cdot (\Theta_0 / \gamma_0)$ in the large- N limit. Since $N \propto \ln M$ (each additional nesting level requires an order-of-magnitude increase in cell count and mass), heart rate $f \propto 1/N$, metabolic rate $B \propto M \cdot f \propto M / \ln M$. This approximates $M^{(3/4)}$ over the typical $\ln M$ range of mammals.

CCT's added value: Weight does not determine lifetime; weight-correlated nesting depth does. This provides the "why" that Kleiber's law empirically describes.

7.2 Evolutionary Senescence Theory and CCT

Evolutionary senescence theory [2, 3] explains aging through declining selection pressure with age — but cannot explain why larger animals evolved longer lifetimes while smaller animals did not. If mutation accumulation were sufficient, smaller animals should have evolved longevity under sufficient selection pressure.

CCT provides a complementary explanation: longevity in large animals is not a "selected advantage" but a forced time-cost of deep nesting under the LEEC conservation constraint. Deep nesting's high maintenance cost forces master rhythm down-regulation; low frequency itself is the direct cause of longevity. Evolution selects the functional advantages of deep nesting (brains, long-distance locomotion); longevity is a structural byproduct.

7.3 Free Energy Principle (FEP)

The FEP [5] offers a unified account of self-organization: systems minimize variational free energy by optimizing the accuracy–complexity trade-off. CCT Θ_0 maps to the FEP generative model's prior confidence; Γ maps to the complexity term accumulation rate; Θ_{crit} maps to the minimum prediction error threshold for self-organization.

However, CCT provides what FEP lacks: temporal-scale dynamics. FEP describes "what the best inference is"; CCT describes "how long the best inference persists." They are complementary: FEP addresses inference space, CCT addresses persistence time.

Crucially, the two frameworks differ in their treatment of suppression: FEP treats emotional suppression as a strategy to minimize prediction error (a form of "action" that changes sensory input). CCT treats it as a malignant increase in internal decoherence rate — suppression may reduce immediate prediction error, but it accelerates the long-term decoherence of the constraint closure, shortening lifetime. This is a key divergence: FEP is synchronic (minimizing current free energy), while CCT is diachronic (tracking cumulative decoherence over time).

7.4 Assembly Theory and CCT

Assembly Theory (AT) [12, 13] defines life through molecular assembly index — the minimum number of steps to assemble a molecule. AT provides a life–nonlife boundary criterion ($A > 15$ for life).

CCT provides a different but orthogonal measure: lifetime as a function of structural depth. An AT-complex molecule with high assembly index that is kinetically unstable would have high A but short T_{life} ; CCT captures temporal persistence where AT captures construction complexity. The two are complementary rather than competing.

7.5 Summary: What CCT Adds

Existing Theory	Domain	What CCT Adds
Kleiber's law	Biology	Provides the structural explanation (nesting depth, not mass)
Senescence theory	Biology	Identifies longevity as structural byproduct, not selected trait
Free Energy Principle	Cognition/Self-organization	Provides temporal persistence dynamics and distinguishes benign vs. malignant suppression
Fracture mechanics	Materials	Provides cross-scale unification across living/nonliving
Assembly Theory	Life – nonlife boundary	Orthogonal: construction vs. persistence

8. Experimentally Testable Predictions

8.1 Prediction 1: Convergent Total Heartbeat Count in Mammals (Benign Condition)

The CCT living branch predicts that, excluding humans (due to modern medical intervention), mammalian total heartbeat counts in benign (non-suppressed) conditions should converge to $10^{(9 \pm 0.5)}$ beats.

Current evidence: All non-human mammals in Table 2 fall within this range [8, 9].

Critical nuance: This prediction holds only for systems in benign deceleration. For systems under chronic suppression or depression, total heartbeat count may be lower while biological aging is accelerated — this is not a falsification but a separate branch

of the theory's prediction (the Γ_{internal} term). In clinical studies, this distinction can be tested by measuring biological age markers (epigenetic clocks, telomere length) alongside heart rate and psychological stress markers.

Potential falsification: If deep-sea mammals — after temperature correction via the Arrhenius factor — show systematic deviations beyond 5×10^8 beats, the theorem would require revision.

Timeframe for testing: 5–10 years, with improving deep-sea biologging technology [10].

8.2 Prediction 2: Ultra-Long Room-Temperature Lifetime for Deep Nesting

($N \geq 6$) Artificial Materials

The non-living branch predicts that covalent structures with $N \geq 6$ (e.g., graphene multilayers, carbon nanotube bundles) should exhibit thermal decoherence lifetimes exceeding 10^9 years at room temperature.

Current evidence: Diamond ($N \approx 6$) shows this lifetime. Other deep-nesting artificial materials have not yet been systematically measured.

Potential falsification: If measurement shows lifetime $< 10^6$ years for clearly $N \geq 6$ structures, the theorem would require revision.

Timeframe for testing: 10–20 years (accelerated aging experiments).

8.3 Prediction 3: Civilizational Knowledge Entropy Rate Criterion

Extending CCT to civilizational scale [6]: $T_{\text{civilization}} = \Theta_{\text{cultural}} / \Gamma_{\text{social}}(N)$. The theorem predicts that when the information generation rate I_{gen} exceeds the purification/compression rate I_{purify} by a ratio $R = I_{\text{gen}} / I_{\text{purify}} > R_c \approx 3$, civilizational coherence time (duration as a recognizable cultural entity) shortens by at least 50%.

Historical case: Late Roman Empire (200–400 CE): documentary proliferation high (I_{gen} high), but cultural N did not increase synchronously — the multi-theistic to mono-theistic transition entailed paradigm confusion. Γ_{social} increased faster than I_{purify} , resulting in the well-documented "overproduction of symbols with loss of meaning anchors."

Potential falsification: If historical analysis shows that high- R civilizations did not experience significant lifespan reduction, the theorem would require revision.

Timeframe: 2–5 years (quantitative historical methodology development).

8.4 Falsification Conditions Summary

Prediction	Falsification Condition	Timeframe
8.1 (benign)	Total heartbeat deviation > 5×10^8 after temperature correction	5 – 10 yrs
8.1 (suppressed)	Accelerated biological aging in low-heart-rate suppressed individuals	5 – 10 yrs
8.2	$N \geq 6$ artificial material lifetime < 10^6 yrs	10 – 20 yrs
8.3	$R > 3$ civilizations show no significant lifespan reduction	2 – 5 yrs

9. Discussion

9.1 Core Contribution

The CCT theorem demonstrates that lifetime is not a biological concept but a universal temporal attribute of constraint closures. The unified criterion $T_{life} = \Theta_0 / \Gamma(N)$ applies across the organic–inorganic divide. Both living systems ("frequency reduction for life extension," $\Gamma_{bio} = \Gamma_0 \cdot e^{(-\alpha N)} + \gamma_0 / N$) and non-living systems ("barrier elevation for lifespan preservation," $\Gamma_{phys} = \Gamma_0 \cdot e^{(-\lambda N)}$) are two symmetric branches of a single ontological constraint: conservation of individual Θ -potential transformation.

9.2 Theoretical Significance

The theorem implies:

- 1. **No ontological discontinuity between living and non-living** — both obey the same temporal dynamics; the difference is only in the functional form of $\Gamma(N)$.
- 2. **Time is not a uniformly flowing river** but the externalization of internal coherence within constraint closures. Deeply nested systems have slower "clocks" (living) or thicker "walls" (non-living) — the universe grants longer existence windows to more deeply nested structures.
- 3. **Heart rate is only a proxy, not the fundamental criterion** — the fundamental criterion is the generalized decoherence rate Γ_{eff} , which includes both physical and internal entropy generation.

9.3 Limitations

Current precision: All cross-scale estimates are B-level (directional criterion, error about ± 1 log unit for lifetimes, ± 2 log units for the $\ln(\Theta_0 / \Theta_{\text{crit}})$ factor). A-level quantification requires independent scale-by-scale calibration of Θ_0 , Θ_{crit} , and Γ_0 .

Environmental fluctuations: The theorem assumes stable T_{env} . Large environmental perturbations (supernovae, asteroid impacts, catastrophic climate shifts) can cause sudden constraint closure decoherence not captured by the model.

Scale-law limitation: Cross-level N comparisons (chemical vs. biological vs. astrophysical) lack normalized scaling-law foundation. This boundary is explicitly stated in Section 6.3.

Distinguishing benign from malignant deceleration: In living systems, empirical separation of Γ_{pump} from Γ_{internal} remains challenging. Biological age markers (epigenetic clocks, inflammatory markers) may provide the needed distinction.

9.4 Ethical and Civilizational Implications

The CCT theorem has implications beyond theoretical physics [6]:

- **Individual longevity:** Lifespan quality is not about maximizing information throughput per unit time, but about sustainable release of the Θ_0 budget within deep nesting structures. Suppression does not extend life — it accelerates internal decoherence.
- **Civilizational persistence:** Civilizational survival depends not on accelerating information production but on increasing the depth of knowledge purification/compression (increasing N of extra-somatic I-structures).

Both are structurally identical to the theorem: reduce decoherence rate, extend coherence time. The universe charges a single account.

10. Conclusion

The Constraint Closure Coherence Time Theorem (CCT) unifies lifetime across living and non-living systems under a single dynamical criterion: $T_{\text{life}} = \Theta_0 / \Gamma(N)$. In living systems, this manifests as frequency reduction for life extension; in non-living systems, as barrier elevation for lifespan preservation. Both are symmetric branches of the same conservation principle — individual Θ -potential transformation is conserved; nesting depth determines its dissipation rate.

Consistent with the rest of the paper. If this is the closing summary or a standalone blurb, the only thing I'd consider is whether "dynamical criterion" is the right phrase given §6.2 explicitly disclaims dynamical isomorphism across scales — "single criterion" or "single conservation criterion" would sidestep the tension.

The theorem provides:

1. A cross-scale framework spanning seven orders of magnitude (molecules to galaxies)
2. Three experimentally testable predictions with specified falsification conditions
3. A unified language for discussing persistence across the organic-inorganic divide
4. A critical distinction between benign and malignant deceleration in living systems
5. Theoretical grounding for civilizational persistence as governed by the same temporal dynamics

The CCT theorem does not replace existing theories but provides the common grammar they lack—a single answer to the question: how long can a structure persist given its internal organization? The universe grants existence not to size, nor to energy, but to coherence. And coherence is not measured by the speed of the pump, but by the integrity of the constraint closure's internal grammar.

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[CCT Theorem — Complete Calibrated Draft]